

Relations and Functions

Relations, functions, Cartesian products via Replacement, isomorphism, quotients
— and the object-theory/metatheory split

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You have used relations (orderings) and functions all your life; the point here is to say what they *are* inside axiomatic set theory, and to notice where an extra axiom (Replacement) is needed to build them.

Relations

Definition (binary relation). R is a (*binary*) *relation* iff R is a set of ordered pairs:

$$\forall u \in R \exists x, y (u = \langle x, y \rangle).$$

Here $\forall u \in R (\dots)$ is shorthand for $\forall u (u \in R \rightarrow \dots)$. We write

$$x R y \quad \text{for} \quad \langle x, y \rangle \in R, \quad \quad x \neg R y \quad \text{for} \quad \langle x, y \rangle \notin R.$$

The abbreviations $x R y$ and $x \neg R y$ are meaningful even when R is not a relation — we will use this later.

Usual terminology

Fix a set A . The following say what it means for R to have a property *on* A ; each is just an abbreviation, obtained by plugging R into a formula.

- **transitive:** $x R y \wedge y R z \rightarrow x R z$
- **reflexive / irreflexive:** $\forall x \in A (x R x) / \forall x \in A (x \neg R x)$
- **trichotomy:** $\forall x, y \in A (x R y \vee y R x \vee x = y)$
- **symmetric:** $x R y \leftrightarrow y R x$

and the three composite notions we care about:

R (strictly) partially orders A R is transitive and irreflexive on A .

R (strictly) totally orders A R is transitive, irreflexive, and satisfies trichotomy on A .

R is an equivalence relation on A R is transitive, reflexive, and symmetric on A .

We take the *strict* $<$ as basic and read $x \leq y$ as $x < y \vee x = y$. For example, $<$ on $\mathbb{Q}$ is a strict total order; $\leq$ is not (it is not irreflexive).

Domain, range, restriction

$$\text{dom}(R) := \{ x : \exists y (\langle x, y \rangle \in R) \}$$

$$\text{ran}(R) := \{ y : \exists x (\langle x, y \rangle \in R) \}$$

$$R \upharpoonright A := \{ \langle x, y \rangle \in R : x \in A \}$$

The definitions are stated for an arbitrary set R , not only for relations. The restriction $R \upharpoonright A$ is used mostly for functions.

Justification (these classes are sets). Recall $\langle x, y \rangle = \{\{x\}, \{x, y\}\}$. So if $\langle x, y \rangle \in R$, then $\{x\}$ and $\{x, y\}$ lie in $\bigcup R$, and hence $x, y \in \bigcup \bigcup R$. Comprehension then gives

$$\text{dom}(R) = \{ x \in \bigcup \bigcup R : \exists y (\langle x, y \rangle \in R) \},$$

and similarly for $\text{ran}(R)$.

Functions

Definition (function). R is a *function* iff R is a relation and for every $x \in \text{dom}(R)$ there is a *unique* y with $\langle x, y \rangle \in R$; that unique y is written $R(x)$.

$F : A \rightarrow B$: F is a function with $\text{dom}(F) = A$ and $\text{ran}(F) \subseteq B$.

$F : A \xrightarrow[\text{onto}]{} B$: $F : A \rightarrow B$ and $\text{ran}(F) = B$ (a *surjection*).

$F : A \xrightarrow{1-1} B$: $F : A \rightarrow B$ and $\forall x, x' \in A (F(x) = F(x') \rightarrow x = x')$ (an *injection*).

$F : A \xrightarrow[\text{onto}]{1-1} B$: both of the above (a *bijection from A onto B*).

Image of a set.

$$F(A) = F''A = \text{ran}(F \upharpoonright A).$$

The notation $F(A)$ is standard but ambiguous when A happens to be both a member and a subset of $\text{dom}(F)$ — e.g. with $\text{dom}(F) = 3 = \{0, 1, 2\}$, is $F(2)$ the value at 2 or the image of $\{0, 1\}$? Write $F''A$ when this could be confused.

Building relations and functions

The axioms used so far (Extensionality, Comprehension, Pairing, Union) do not let us build many relations. The obvious source is the **Cartesian product**

$$A \times B = \{ \langle x, y \rangle : x \in A \wedge y \in B \},$$

after which any relation between A and B is a subset of $A \times B$, cut out by Comprehension. So everything reduces to: *does $A \times B$ exist?*

First method: Power Set

If $x \in A$ and $y \in B$ then $\{x\}, \{x, y\} \subseteq A \cup B$, so $\langle x, y \rangle = \{\{x\}, \{x, y\}\} \in \mathcal{P}(\mathcal{P}(A \cup B))$. Hence

$$A \times B = \{ w \in \mathcal{P}(\mathcal{P}(A \cup B)) : \exists x \in A \exists y \in B (w = \langle x, y \rangle) \}$$

exists by Power Set (twice) and Comprehension.

This works, but it is a little “wasteful” in terms of *cardinality*. The sets

$$\{0\} \times \omega, \quad \text{id}_\omega = \{ \langle x, x \rangle : x \in \omega \}$$

have the same “size” as ω , yet the justification above manufactures them as subsets of $\mathcal{P}(\mathcal{P}(\omega))$ — a set of size $2^{2^{\aleph_0}}$. We are invoking the existence of a huge set to produce a countable one.

Second method: Replacement

General method. Suppose $\forall x \in A \exists! y \varphi(x, y)$. Then Replacement gives a set B with $\forall x \in A \exists y \in B \varphi(x, y)$, and Comprehension trims it to $\{ y \in B : \exists x \in A \varphi(x, y) \} = \{ y : \exists x \in A \varphi(x, y) \}$. So *the image of a set under a definable single-valued relation is a set*.

Apply this to build $A \times B$ without Power Set. Fix $b \in B$ and let

$$\varphi(x, y) \equiv y = \langle x, b \rangle.$$

Then $\forall x \in A \exists! y \varphi(x, y)$, so by Replacement + Comprehension the set

$$A \times \{b\} = \{ \langle x, b \rangle : x \in A \}$$

exists. Applying Replacement again — now over $b \in B$ — gives

$$S := \{ A \times \{b\} : b \in B \},$$

and finally $A \times B = \bigcup S$ by Union.

This argument never opened up $\langle x, b \rangle$; it used only that $\langle x, b \rangle$ is *some* set determined by x and b . So it proves $A \times B$ (and also $\text{dom}(R)$, $\text{ran}(R)$) exist for **any** definition of the ordered pair.

Packaging the general method as a statement about functions:

Lemma. If $\forall x \in A \exists! y \varphi(x, y)$, then there is a function f with $\text{dom}(f) = A$ such that for each $x \in A$, $f(x)$ is the unique y with $\varphi(x, y)$.

Proof. Fix B from Replacement and set $f = \{ \langle x, y \rangle \in A \times B : \varphi(x, y) \}$. $\square$

Inverses, composition, isomorphism

$R^{-1} := \{ \langle y, x \rangle : \langle x, y \rangle \in R \}$ a subset of $\text{ran}(R) \times \text{dom}(R)$. If f is a function, f^{-1} is a function iff f is 1-1.

$G \circ F := \{ \langle x, z \rangle : \exists y (\langle x, y \rangle \in F \wedge \langle y, z \rangle \in G) \}$ when F, G are functions with $\text{ran}(F) \subseteq \text{dom}(G)$, this is just $(G \circ F)(x) = G(F(x))$.

Isomorphism. Let R be a binary relation on A and S a binary relation on B . Then F is an *isomorphism* from (A, R) onto (B, S) iff

$$F : A \xrightarrow[\text{onto}]{1-1} B \quad \text{and} \quad \forall x, y \in A [x R y \leftrightarrow F(x) S F(y)].$$

We write $(A, R) \cong (B, S)$ when such an F exists. The definition makes sense for arbitrary relations; for now we use it for orderings.

Equivalence relations and quotients

Let R be an equivalence relation on a set A , and $x \in A$. Define

$$[x] := \{ y \in A : x R y \} \quad (\text{Comprehension}),$$

the *equivalence class* of x , and

$$A/R := \{ [x] : x \in A \}.$$

To see A/R is a set, apply the general method with $\varphi(x, y) \equiv y = [x]$: by Lemma I.7.9 there is a function f with $\text{dom}(f) = A$ and $f(x) = [x]$ — the **quotient map** $x \mapsto [x]$ — and then $A/R = \text{ran}(f)$.

This is the standard justification, in books on group theory and topology, for forming a quotient of a structure by an equivalence relation; the extra structure on A/R is defined separately.

Object theory versus metatheory

The informal notions “relation” and “function” get **two** representations in set theory: as *sets* (objects of the formal theory ZFC), and as *abbreviations* in the metatheory.

As sets

A relation is a specific kind of set, handled inside ZFC. The defining property is a formula of the object theory — for example

$$\text{Rel}(R) \equiv \forall u \in R \exists y, z (u = \langle y, z \rangle).$$

We then prove facts *about* such sets as theorems of ZFC, e.g.

$$\text{ZFC} \vdash \text{Rel}(R) \rightarrow \exists z (z = \text{dom}(R)),$$

using the justification given earlier. A concrete function such as $f = \{\langle 1, 2 \rangle, \langle 2, 2 \rangle, \langle 3, 1 \rangle\}$ is likewise a set formally defined in ZFC.

As abbreviations in the metatheory

By contrast, $\in$, $=$, $\subseteq$ are *informal* relations — a different kind of animal. The “relation”

$$S = \{p : \exists x, y (p = \langle x, y \rangle \wedge x \subseteq y)\}$$

does **not** exist: it is a proper class, since $\text{dom}(S)$ would be the universe V . We view $x \subseteq y$ instead as an abbreviation in the metatheory for $\forall z (z \in x \rightarrow z \in y)$; likewise $(A, R) \cong (B, S)$ abbreviates a longer statement. Membership and equality, $\in$ and $=$, are already primitive symbols.

Many properties from the terminology list still make sense for such **pseudorelations**, by plugging them into the defining formula. For instance $\in$ totally orders $3 = \{0, 1, 2\}$ (i.e. $\forall x, y, z \in$

$3[x \in y \wedge y \in z \rightarrow x \in z]$, plus trichotomy), while $\subseteq$ is transitive on all of V but not a total order of 3, and $\in$ is not transitive on V (since $0 \in \{0\} \in \{\{0\}\}$ but $0 \notin \{\{0\}\}$).

Similarly for functions: $\bigcup : V \rightarrow V$ and $\mathcal{P} : V \rightarrow V$ are **pseudofunctions**. There is no set of pairs $\bigcup = \{\langle \mathcal{F}, \bigcup \mathcal{F} \rangle : \mathcal{F} \in V\}$; instead there is a formula $\varphi(\mathcal{F}, Z)$ saying “ Z is the union of the members of $\mathcal{F}$ ”, we prove $\forall \mathcal{F} \exists! Z \varphi(\mathcal{F}, Z)$, and $\bigcup \mathcal{F}$ *denotes* that Z . Function properties still apply as abbreviations — e.g. “ $\bigcup$ is not 1-1” is the formula

$$\exists x_1, x_2, y [\varphi(x_1, y) \wedge \varphi(x_2, y) \wedge x_1 \neq x_2].$$